

AlmaLinux OS 9.5, a free alternative to RHEL 9.5, unveiled

The AlmaLinux OS Foundation has unveiled AlmaLinux OS 9.5, codenamed ‘Teal Serval’, offering the open source community a powerful and free alternative to Red Hat Enterprise Linux (RHEL) 9.5. Built to mirror RHEL’s capabilities, AlmaLinux OS 9.5 combines stability with features to enhance system performance and developer productivity.

This release introduces significant updates for system monitoring, performance visualisation, and data collection. It also includes improvements in security, with revamped cryptographic policies that now extend to Java algorithm selection, fortifying the platform against emerging threats.

For developers, AlmaLinux OS 9.5 delivers new and updated compilers such as GCC 14, LLVM 18.1.8, Rust 1.79.0, and Go 1.22, enhancing application development and management. Additionally, SELinux policies have been refined to allow the QEMU Guest Agent to execute confined commands, while ca-certificates have been updated to better support trusted CA roots in the OpenSSL directory format.

The release also brings an array of updated components, including .NET 9.0, Node.js 22, Apache 2.4.62, and more, ensuring compatibility with modern application frameworks. New tools such as BIND 9.18, GDB 14.2, and Grafana 10.2.6 further expand its utility for developers and systems administrators alike.

Powered by Linux kernel 5.14.0-427.13.1.el9_4, AlmaLinux OS 9.5 offers robust support for various architectures, including x86_64, AArch64, PowerPC 64-bit Little Endian, and IBM system Z, ensuring compatibility across a wide range of systems.

Users seeking to explore AlmaLinux OS 9.5 can download installation ISOs from the official website. Current users can easily upgrade to the latest version using a single command in their terminal.

the next wave of innovation tomorrow,” said Joe Fernandes, vice president and general manager, AI Business Unit, Red Hat.

As businesses increasingly shift to AI-driven practices, the need for AI-ready infrastructure spanning applications, data, platforms, and hybrid environments has never been more critical. RHEL AI on Microsoft Azure addresses these needs



by offering a consistent and scalable platform for building, testing, and deploying gen AI models across hybrid cloud setups.

RHEL AI powers generative AI innovation with enterprise-grade, open source-licensed Granite models and supports model alignment to private enterprise data through InstructLab tooling. Furthermore, it provides an efficient pathway to Red Hat OpenShift AI, enabling organisations to finetune and scale models using familiar tools and concepts.

RHEL AI is the latest addition to Red Hat’s AI-optimised offerings on Azure, joining solutions like Red Hat OpenShift AI and Azure Red Hat OpenShift—a fully managed application platform co-developed by the two companies.

With RHEL AI on Azure, organisations gain greater flexibility and a robust foundation to drive generative AI innovation, making it easier to achieve their strategic goals in an AI-driven era.

Jake Zborowski, general manager, Microsoft Azure Platform, Microsoft Corp., said: “We’re pleased to welcome Red Hat Enterprise Linux AI to the Microsoft Azure Marketplace, which gives our partners great exposure to cloud customers around the globe. Azure Marketplace offers world-class quality experiences from global trusted partners with solutions tested to work seamlessly with Azure.”

Mcicity digital twin unveiled for connected and autonomous vehicle testing

The University of Michigan’s Mcicity Test Facility has unveiled the first open source digital twin of its renowned test track, enabling researchers worldwide to simulate and test connected and autonomous vehicle (CAV) technologies. This groundbreaking tool, freely accessible, offers a safer, faster, and cost-effective alternative to real-world testing.

A digital twin is a virtual replica of a physical environment that interacts with real-world data, enabling advanced simulations. Researchers can now virtually access Mcicity’s comprehensive testing environment, featuring diverse road materials, markings, intersections, and traffic signals, without travelling to the test track.

The digital twin integrates with TeraSim, Mcicity’s open source traffic simulator, to create dynamic testing scenarios. It introduces pedestrians, cyclists, and vehicles while generating safety-critical events, such as potential collisions. Using traffic models calibrated with real-world data, it simulates both every day and high-risk driving scenarios.

Experts view CAV technologies as key to safer, more efficient transportation, though substantial research is still required to achieve widespread benefits. Virtual testing accelerates progress, enabling autonomous vehicles to log “millions of miles” in simulations before real-world deployment. “With the digital twin, we control variables like simulated pedestrian traffic, focusing and accelerating testing,” said Mcicity software engineer Darian Hogue.